

- Web-based materials/chemistry simulation platform

Materials Square Class Curriculum

○



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[Lab.3] Adsorption energies of Li atoms for different coverage	 <u>18</u>
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[Lab.3] Adsorption energies of Li atoms for different coverage

Learning Objectives:

→ To investigate the lithium storage capacity of layer structures, calculate the adsorption energy differences depending on the adsorption sites and number of lithium atoms on the 2D material.

Materials: Li/MoS₂



MatSQ Credit: ~ 360 CPU hour (\$ 90)

[Lab.3] Adsorption energies of Li atoms for different coverage

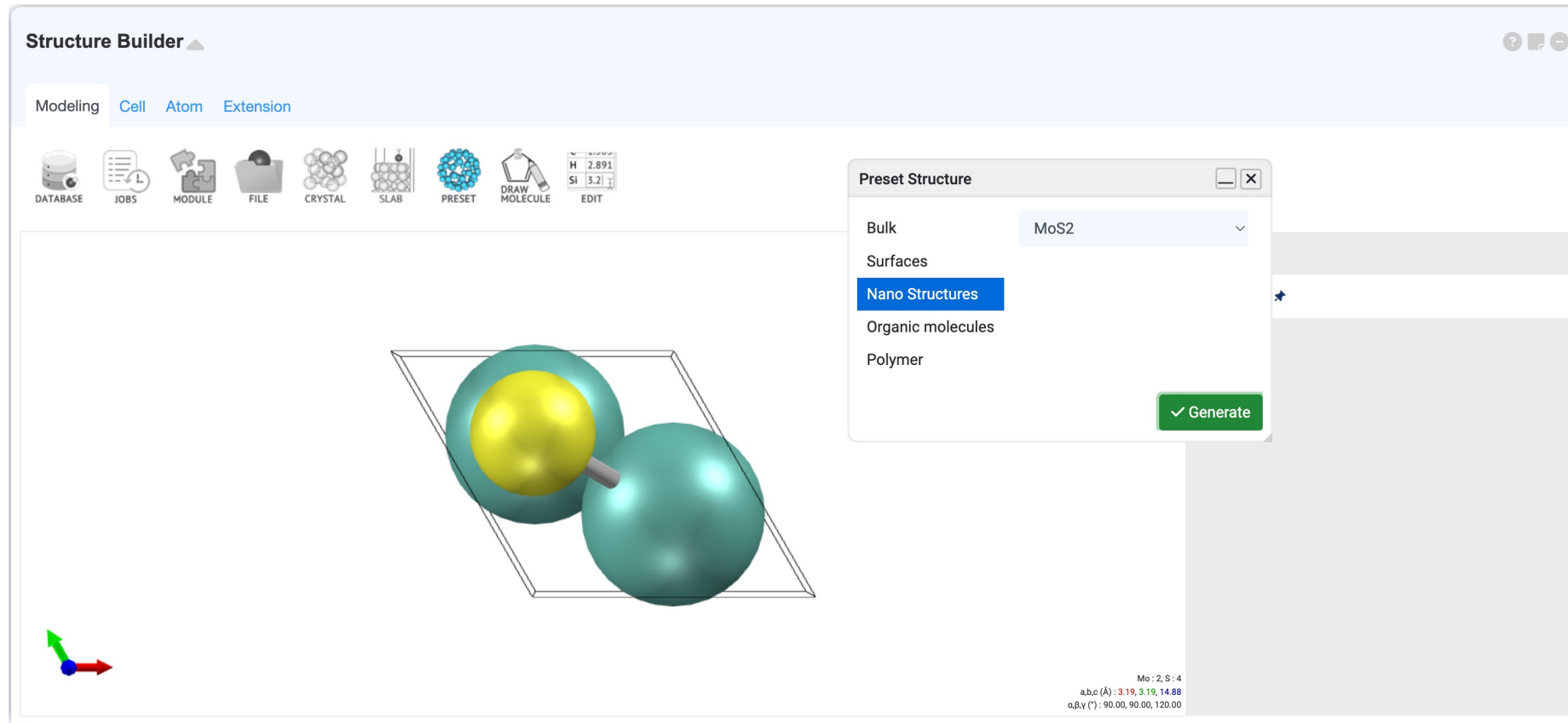
Li/MoS₂ layer

1. Build 2h-MoS₂ layer
2. Calculate variable cell relaxation of primitive cell
3. Build MoS₂ supercell with optimized lattice constant
4. Calculate MoS₂ supercell energy
5. Calculate Li atomic energy
6. Build adsorption structures
7. Calculate atomic structure optimization (Li-atoms/slab)
8. Calculate adsorption energies
9. Build adsorption structures of Li atoms for different coverage
10. Calculate atomic structure optimization (Li-atoms/slab)
11. Calculate adsorption energies

MatSQ Credit: ~ 360 CPU hour (\$ 90)

[Lab.3] Adsorption energies of Li atoms for different coverage

1. Build 2h-MoS₂ layer



[Lab.3] Adsorption energies of Li atoms for different coverage

1. Build 2h-MoS₂ layer

Structure Builder

Modeling Cell Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular Normal

Manual Edit

Cell Lattice Vectors

a	3.19032	0	0
b	-1.595155	2.76289	0
c	0	0	20

Atoms Cartesian ☐ Selected Atoms

ID	EL.	x	y	z
1	Mo	0.00000	1.84192	8.71975
2	S	1.59516	0.92096	10.2845
3	S	1.59516	0.92096	7.15492

+ Add Atom

Reset Apply

[Lab.3] Adsorption energies of Li atoms for different coverage

2. Calculate variable cell relaxation of primitive cell

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info Update Structure Keyword information

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure

High (long time)

Isolate System

Ionic+cell optimization

☐ Spin polarization

☐ Van der Waals

☐ Electric field

☐ Spin Orbit Coupling

☐ DFT+U

&CONTROL

Calculation type

Max SCF steps

Information amount

Force threshold (Ry/bohr)

vc-relax

200

high

0.00038

&SYSTEM

E_{cut}(wfc)(Ry)

E_{cut}(rho)(Ry)

Occupations

Gaussian broadening (Ry)

30

180

smearing

0.002

&ELECTRONS

Max iteration step

SCF must converge

Mixing mode

Mixing beta

Convergence threshold (Ry)

Starting wavefunction

200

True

plain

0.5

0.000001

random

&IONS

Ion dynamics

Upscale

bfgs

100

&CELL

Cell dynamics

Cell factor

Pressure threshold (Kbar)

Cell relax type

bfgs

2

0.2

2Dxy

KPOINTS

Sampling

Grid

Shift

Monkhorst-Pack

991

000

[Lab.3] Adsorption energies of Li atoms for different coverage

3. Build MoS₂ supercell with optimized lattice constant

Structure Builder

Modeling Cell Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Manual Edit

Cell Lattice Parameters

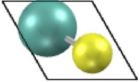
a	3.17254088500	α (°)	90
b	3.17254088500	β (°)	90
c	20	γ (°)	119.999999999

Atoms Fractional ☐ Selected Atoms

ID	EL.	a	b	c
1	Mo	0.33333	0.66666	0.43598
2	S	0.66666	0.33333	0.51430
3	S	0.66666	0.33333	0.35766

+ Add Atom

Reset Apply



[Lab.3] Adsorption energies of Li atoms for different coverage

3. Build MoS₂ supercell with optimized lattice constant

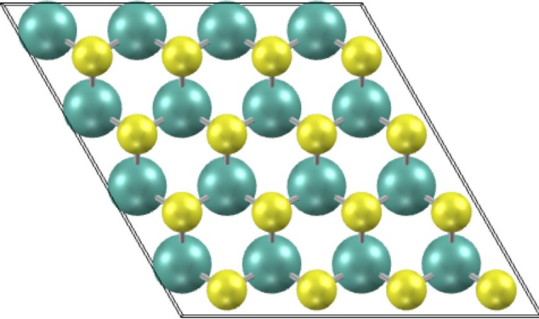
Structure Builder

Modeling Cell Atom Extension

STRAIN CLONE VACUUM CLEAVE MERGE PRIMITIVE CONVENTIONAL

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular
A U B



Clone (# of Cell)

☒ Clone with cell ☐ Paste selected

a	b	c
4	4	1

☐ Clone selected atom(s)

✓ Apply

Mo : 16, S : 32
a,b,c (Å) : 12.69, 12.69, 20.00
α,β,γ (°) : 90.00, 90.00, 120.00

[Lab.3] Adsorption energies of Li atoms for different coverage

4. Calculate total energy of MoS₂ supercell

Quantum Espresso

PWscf + DOS ▼

Scripting Option : Template ▼ Atom info Update Structure Keyword information

Target property
Energy / Structure ▼

Precision
High (long time) ▼

Model type
Isolate System ▼

Structure optimization
No optimization ▼

Option
☐ Spin polarization
☐ Van der Waals
☐ Electric field
☐ Spin Orbit Coupling
☐ DFT+U

&CONTROL
Calculation type
Information amount
Force threshold (Ry/bohr)

scf ▼
high ▼
0.001

&SYSTEM
E_{cut}(wfc)(Ry)
E_{cut}(rho)(Ry)
Occupations
Gaussian broadening (Ry)

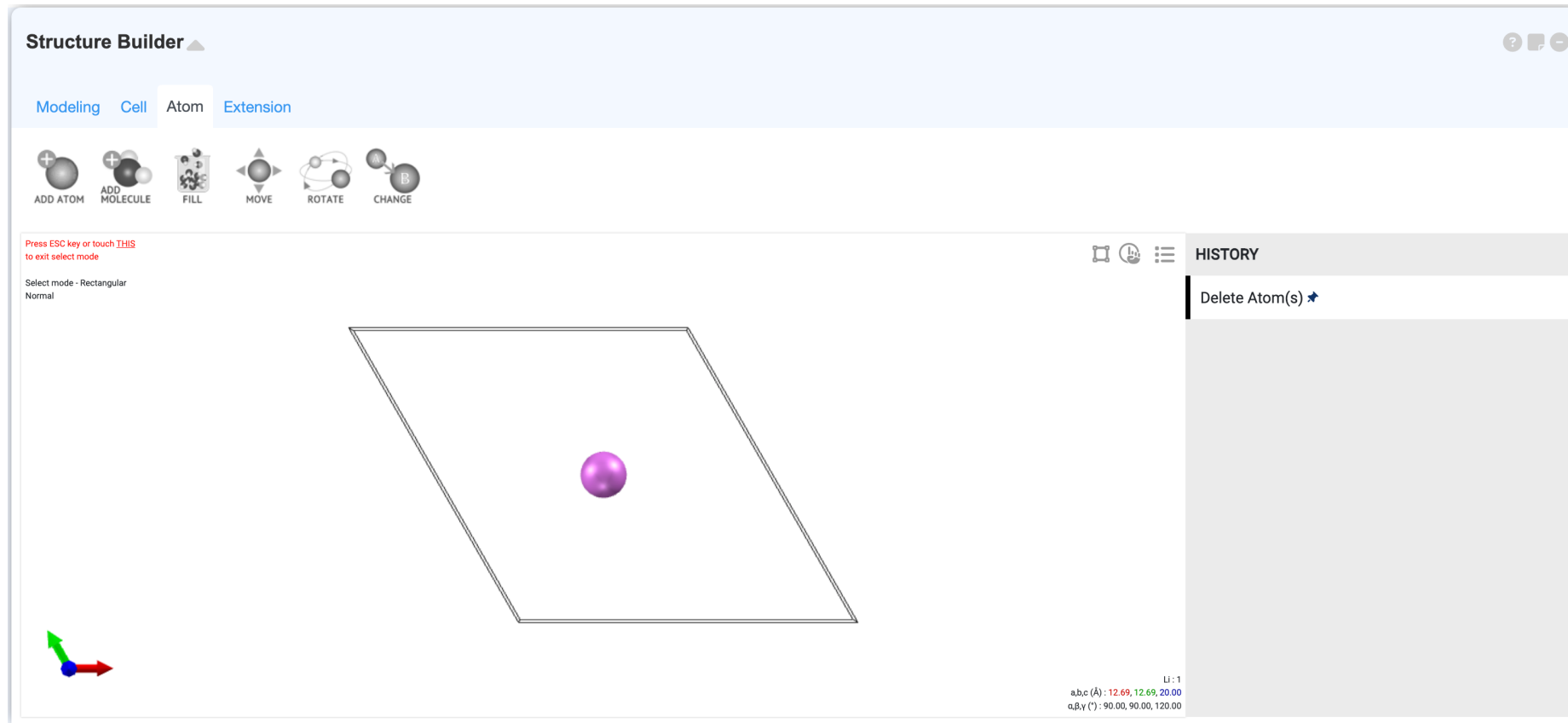
30
180
smearing ▼
0.002

&ELECTRONS
Max iteration step
SCF must converge
Mixing mode
Mixing beta
Convergence threshold (Ry)
Starting wavefunction
KPOINTS
Sampling
Grid
Shift

200
True ▼
plain ▼
0.5
0.000001
random ▼
Monkhorst-Pack ▼
2 2 1
0 0 0

[Lab.3] Adsorption energies of Li atoms for different coverage

5. Calculate Li atomic energy



[Lab.3] Adsorption energies of Li atoms for different coverage

5. Calculate Li atomic energy

Quantum Espresso    

PWscf  DOS 

Scripting Option : Template   Atom info ☒ Update Structure  Keyword information

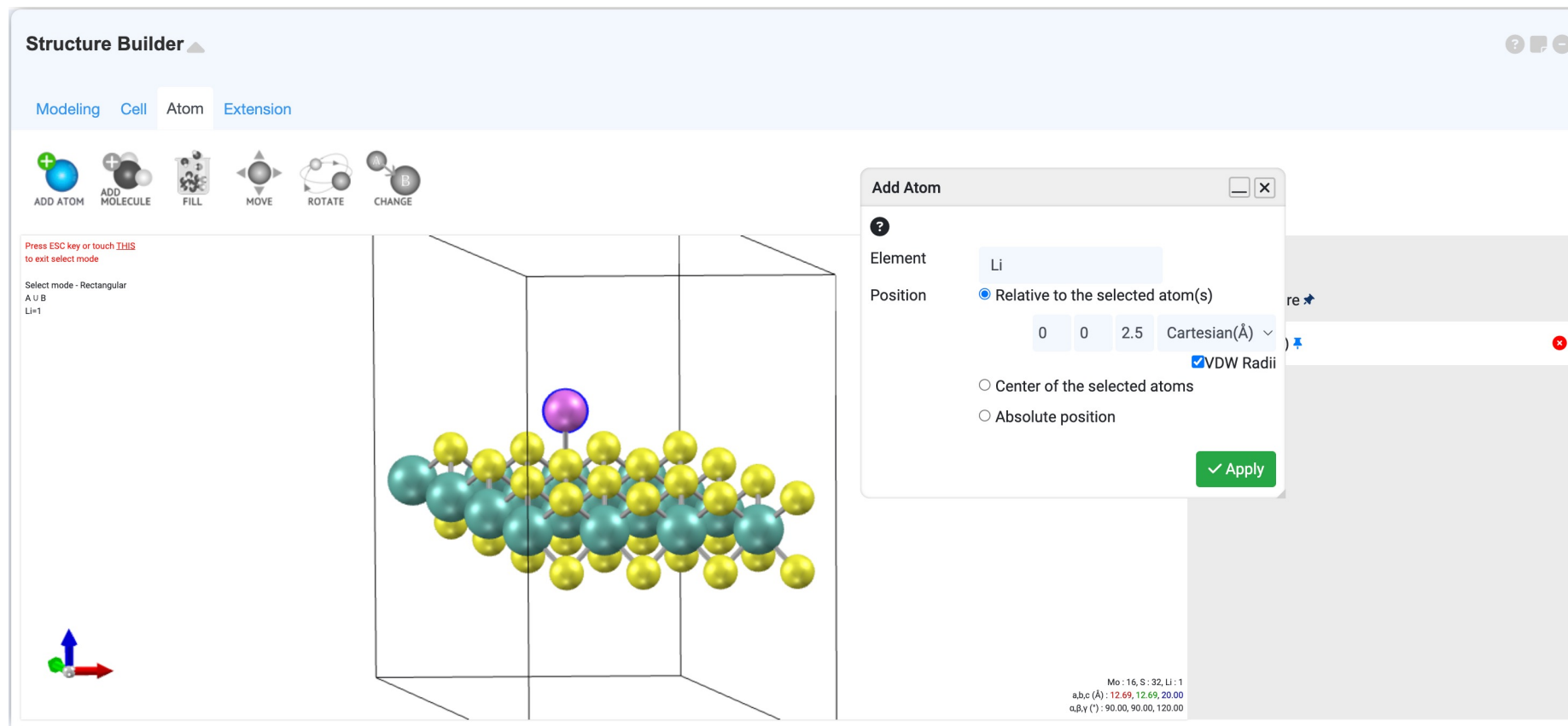
Target property: Energy / Structure  Precision: High (long time)  Model type: Molecule / single atom  Structure optimization: No optimization  Option:

- ☒ Spin polarization
- ☐ Van der Waals
- ☐ Electric field
- ☐ Spin Orbit Coupling
- ☐ DFT+U

&CONTROL		&SYSTEM		&ELECTRONS	
Calculation type	scf 	$E_{\text{cut}}(\text{wfc})(\text{Ry})$	30	Max iteration step	200
Information amount	high 	$E_{\text{cut}}(\text{rho})(\text{Ry})$	180	SCF must converge	True 
Force threshold (Ry/bohr)	0.00038	Occupations	smearing 	Mixing mode	plain 
		Gaussian broadening (Ry)	0.002	Mixing beta	0.5
		Number of electron spin	2 (up,down) 	Convergence threshold (Ry)	0.000001
		Starting magnetization		Starting wavefunction	random 
		Li	1		
				KPOINTS	
				Sampling	Monkhorst-Pack 
				Grid	1 1 1
				Shift	0 0 0

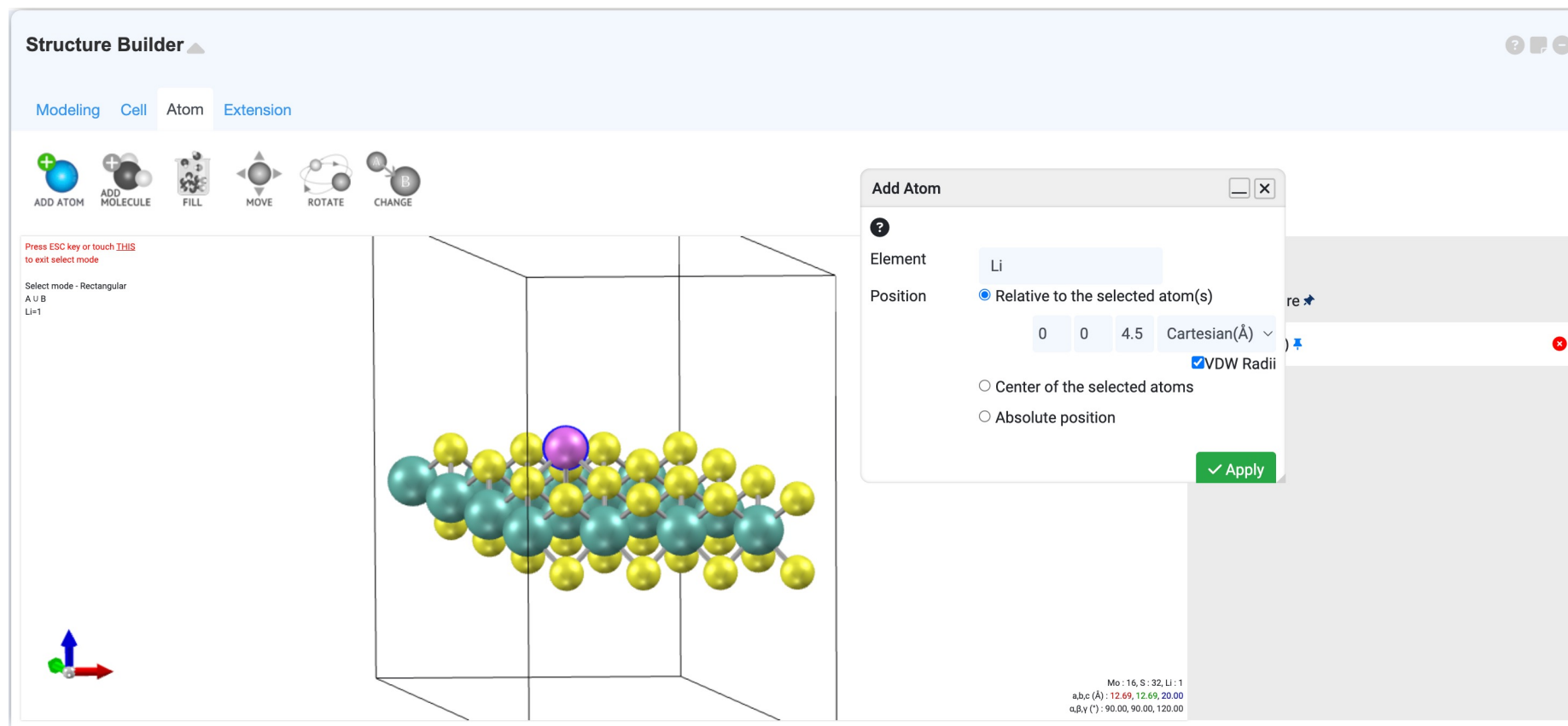
[Lab.3] Adsorption energies of Li atoms for different coverage

6. Build adsorption structures



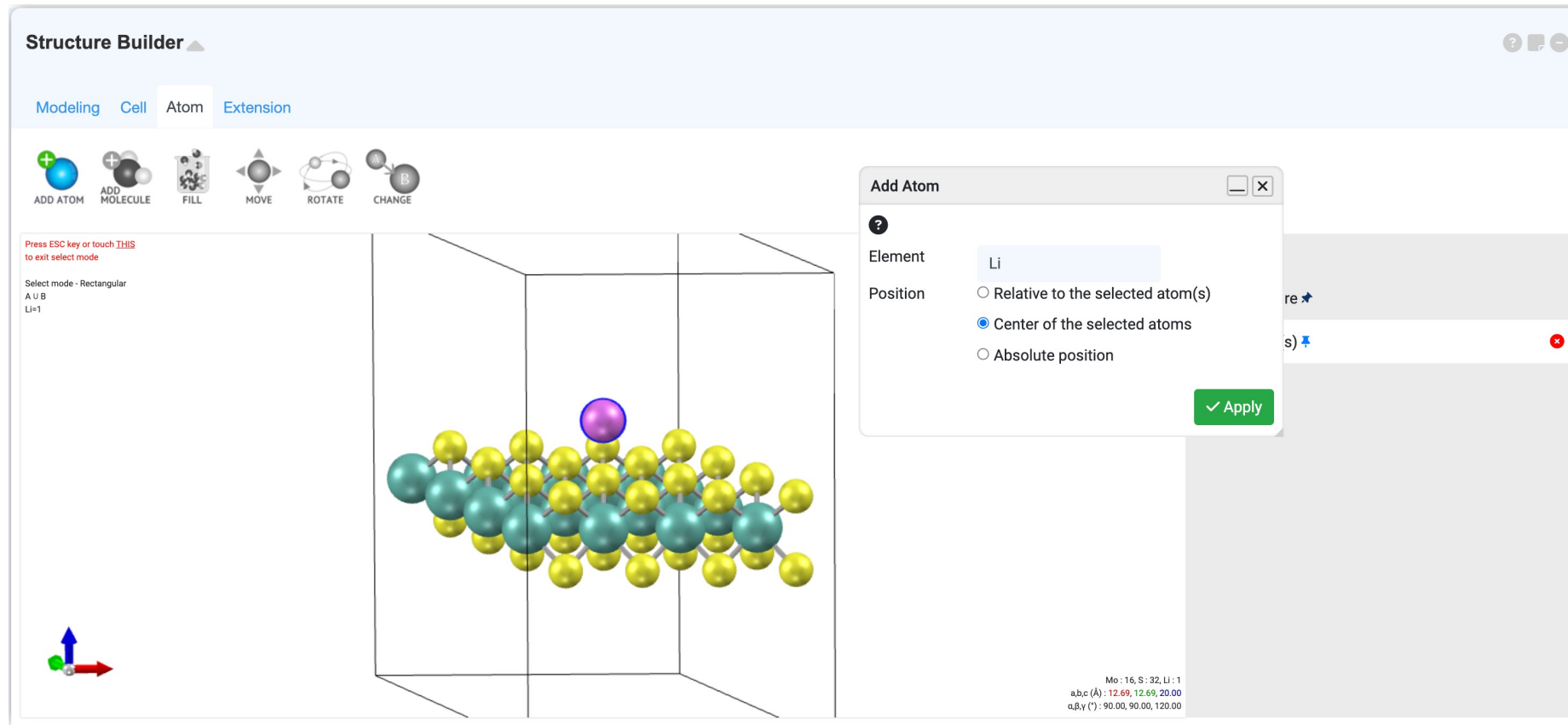
[Lab.3] Adsorption energies of Li atoms for different coverage

6. Build adsorption structures



[Lab.3] Adsorption energies of Li atoms for different coverage

6. Build adsorption structures



[Lab.3] Adsorption energies of Li atoms for different coverage

7. Calculate atomic structure optimization

Quantum Espresso

PWscf + DOS ▼

Scripting Option : Template ▼ [Atom info](#) ☒ Update Structure [Keyword information](#)

Target property
Energy / Structure ▼

Precision
High (long time) ▼

Model type ?
Isolate System ▼

Structure optimization
Ionic optimization ▼

Option
☐ Spin polarization
☐ Van der Waals
☐ Electric field
☐ Spin Orbit Coupling
☐ DFT+U

&CONTROL
Calculation type relax ▼
Max SCF steps 200
Information amount high ▼
Force threshold (Ry/bohr) 0.001

&SYSTEM
 $E_{\text{cut}}(\text{wfc})(\text{Ry})$ 30
 $E_{\text{cut}}(\text{rho})(\text{Ry})$ 180
Occupations smearing ▼
Gaussian broadening (Ry) 0.002

&ELECTRONS
Max iteration step 200
SCF must converge True ▼
Mixing mode plain ▼
Mixing beta 0.5
Convergence threshold (Ry) 0.000001
Starting wavefunction random ▼

KPOINTS
Sampling Monkhorst-Pack ▼
Grid 2 2 1
Shift 0 0 0

&IONS
Ion dynamics bfgs ▼
Upscale 100

[Lab.3] Adsorption energies of Li atoms for different coverage

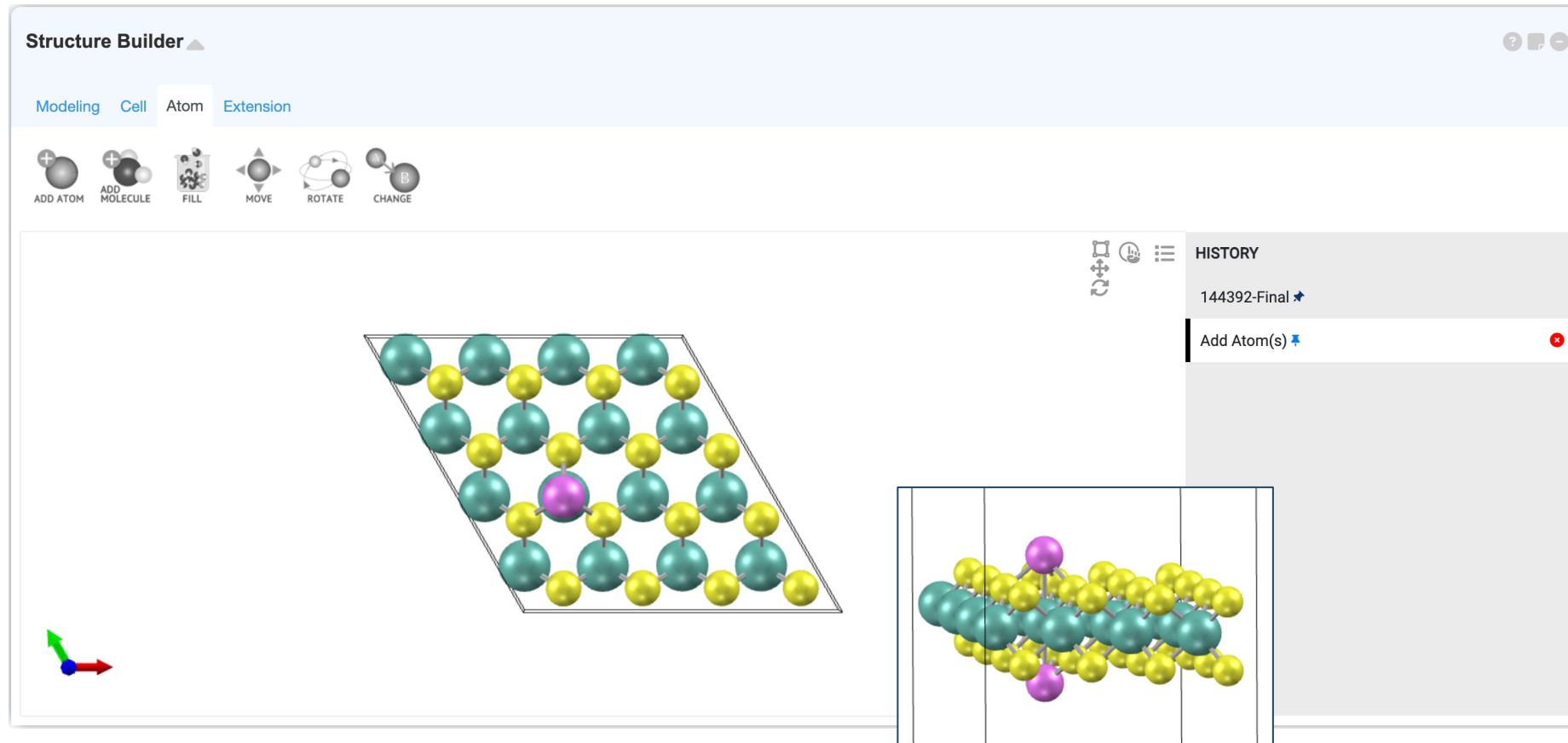
8. Calculate adsorption energies

$$E_{\text{adsorption}} = -[E_{\text{adsorption structure}} - (E_{\text{Li-atom}} + E_{\text{MoS}_2})]$$



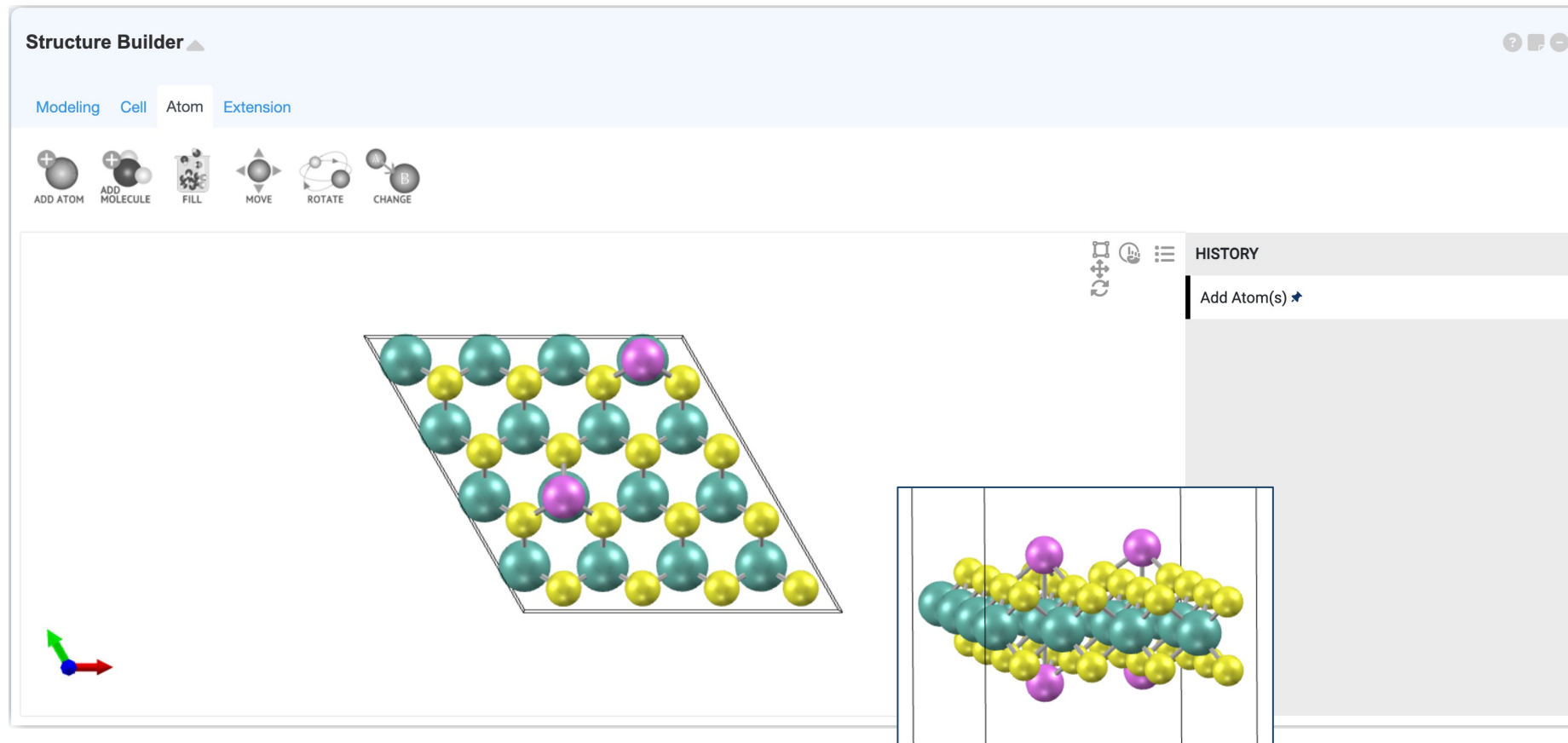
[Lab.3] Adsorption energies of Li atoms for different coverage

9. Build adsorption structures of Li atoms for different coverage



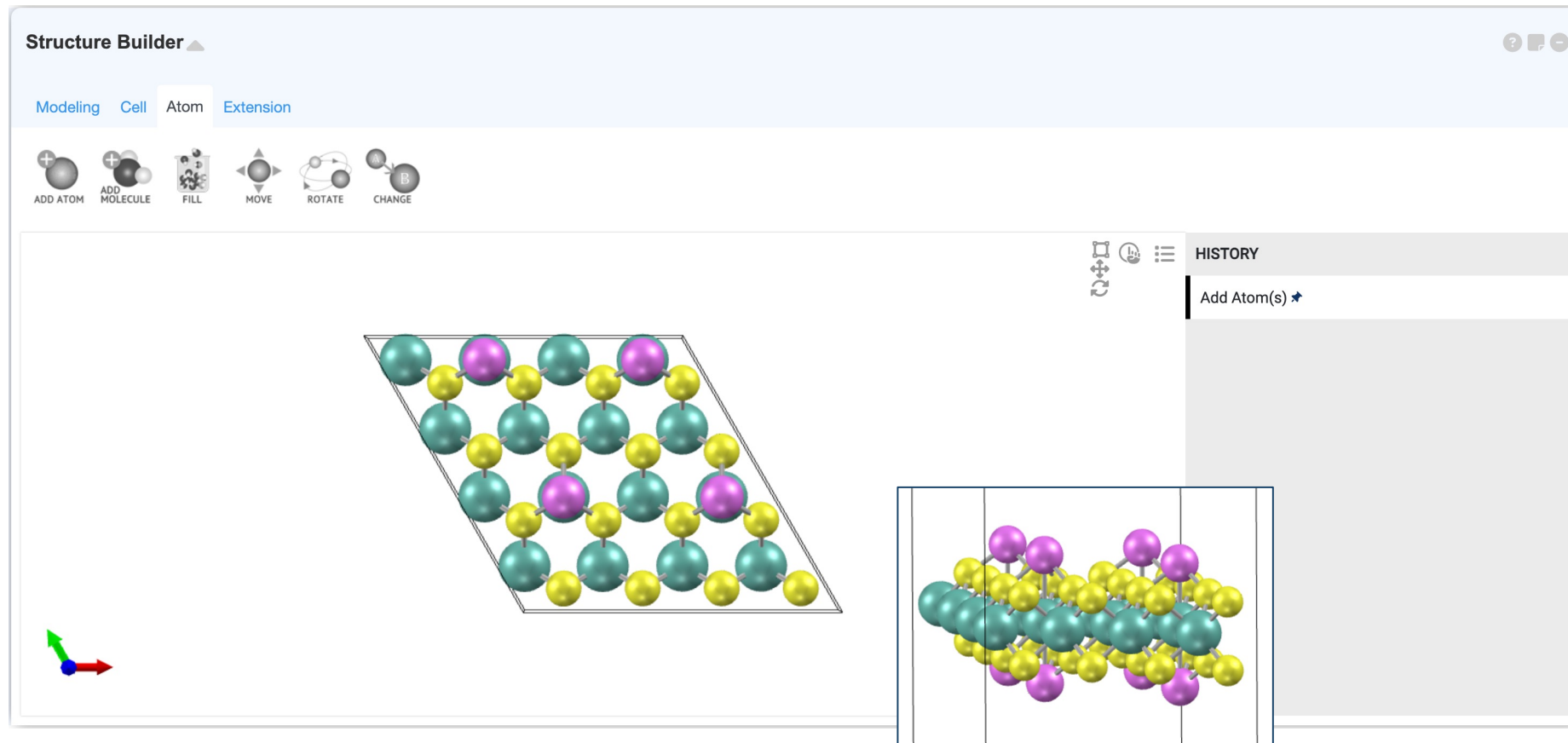
[Lab.3] Adsorption energies of Li atoms for different coverage

9. Build adsorption structures of Li atoms for different coverage

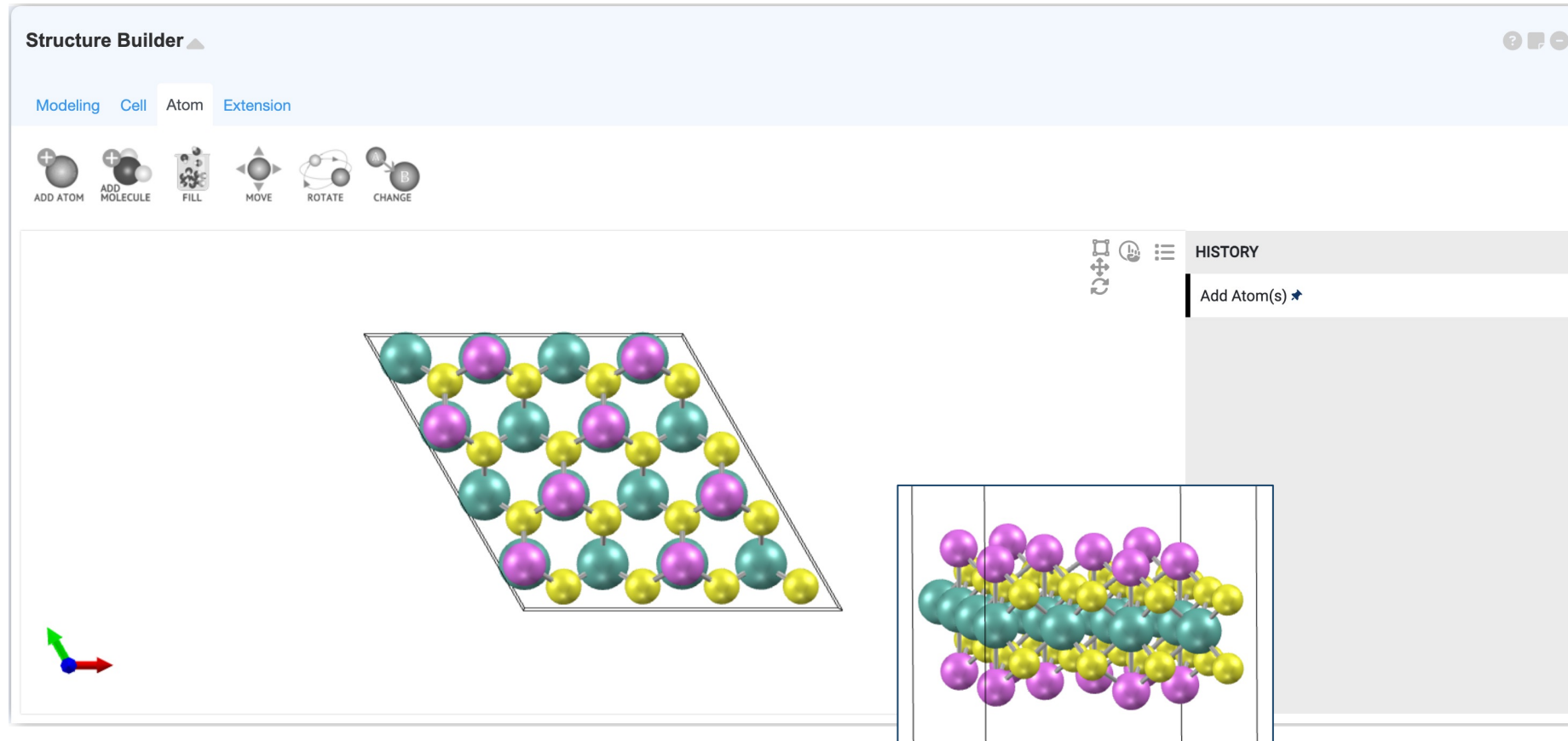


[Lab.3] Adsorption energies of Li atoms for different coverage

9. Build adsorption structures of Li atoms for different coverage

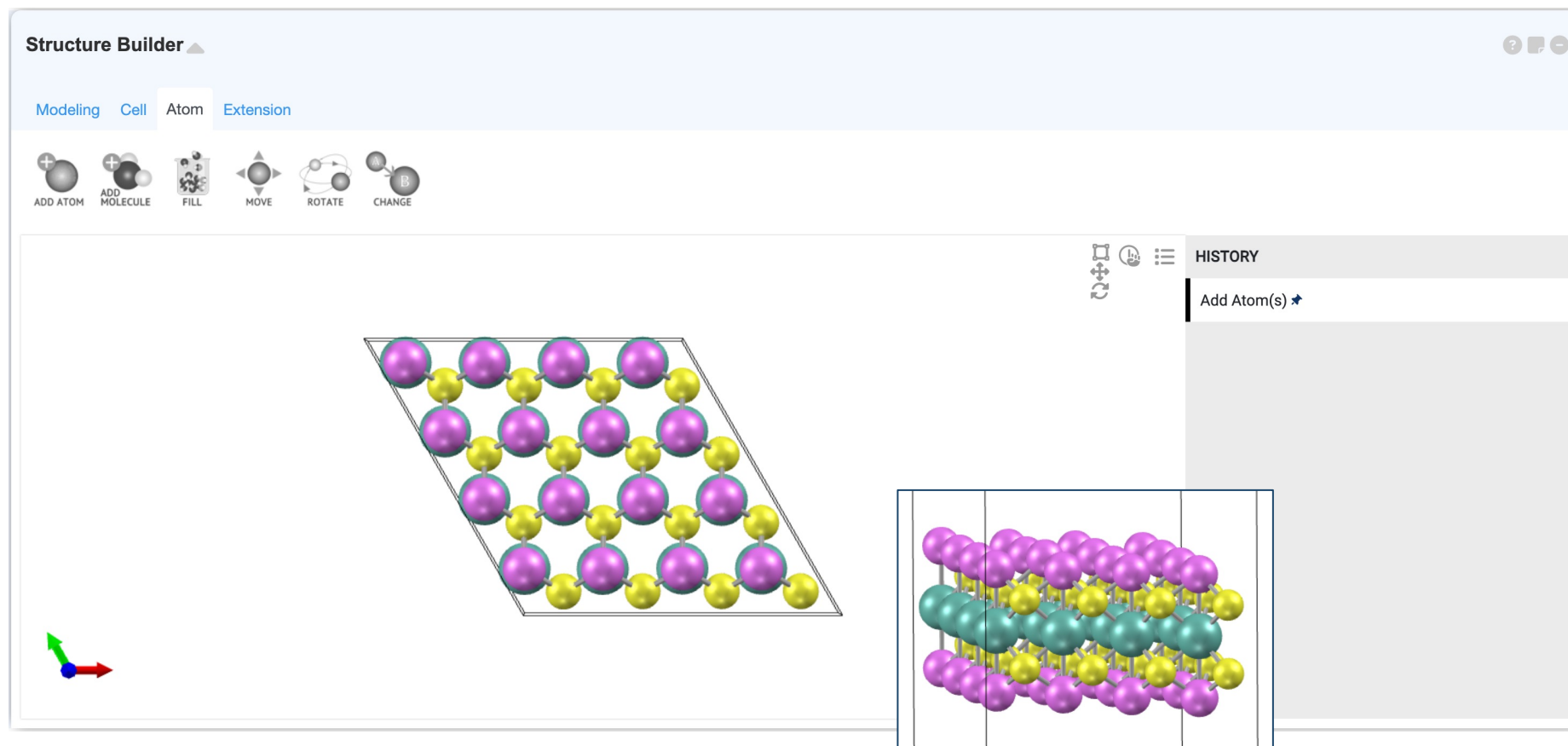


[Lab.3] Adsorption energies of Li atoms for different coverage



[Lab.3] Adsorption energies of Li atoms for different coverage

9. Build adsorption structures of Li atoms for different coverage



[Lab.3] Adsorption energies of Li atoms for different coverage

10. Calculate atomic structure optimization

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info Update Structure Keyword information

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure

High (long time)

Isolate System

Ionic optimization

☐ Spin polarization
☐ Van der Waals
☐ Electric field
☐ Spin Orbit Coupling
☐ DFT+U

&CONTROL

Calculation type

Max SCF steps

Information amount

Force threshold (Ry/bohr)

&SYSTEM

$E_{\text{cut}}(\text{wfc})(\text{Ry})$

$E_{\text{cut}}(\text{rho})(\text{Ry})$

Occupations

Gaussian broadening (Ry)

&ELECTRONS

Max iteration step

SCF must converge

Mixing mode

Mixing beta

Convergence threshold (Ry)

Starting wavefunction

KPOINTS

Sampling

Grid

Shift

&IONS

Ion dynamics

Upscale

relax

200

high

0.001

30

180

smearing

0.002

200

True

plain

0.5

0.000001

random

Monkhorst-Pack

2 2 1

0 0 0

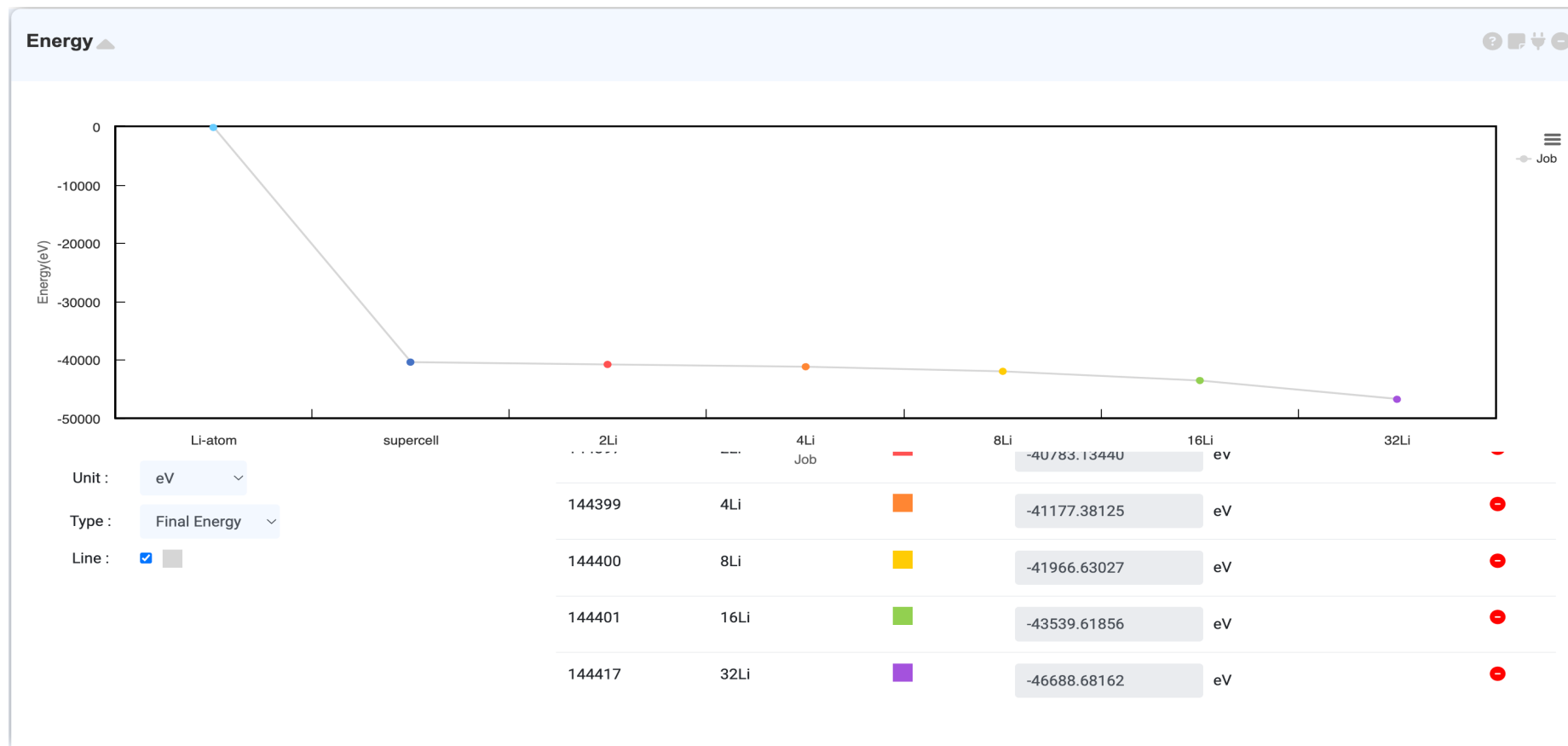
bfgs

100

[Lab.3] Adsorption energies of Li atoms for different coverage

11. Calculate adsorption energies

$$E_{\text{adsorption}} = -[E_{\text{adsorption structure}} - (n_{\text{Li-atoms}} \times E_{\text{Li-atom}} + E_{\text{MoS}_2})] / n_{\text{Li-atoms}}$$



[Lab.3] Adsorption energies of Li atoms for different coverage

11. Calculate adsorption energies

$$E_{\text{adsorption}} = -[E_{\text{adsorption structure}} - (n_{\text{Li-atoms}} \times E_{\text{Li-atom}} + E_{\text{MoS}_2})] / n_{\text{Li-atoms}}$$

Memo ▲

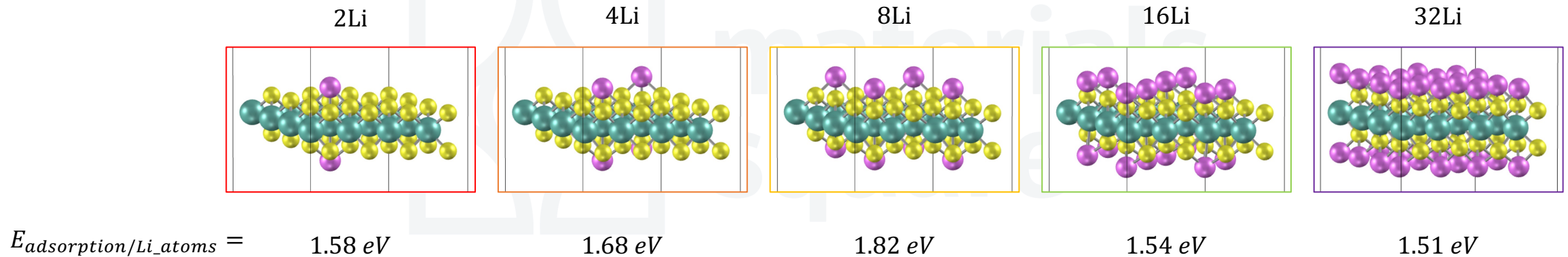
Note Sheet

	A ▼	B ▼	C ▼	D ▼	E ▼	F ▼	G ▼	H ▼	I ▼	J ▼	K ▼	L ▼
1		# of Li atoms	Total Energy	Adsorption Energy								
2	Li atom		-195.34742									
3	MoS2		-40389.26570									
4	2Li/	2	-40783.13440	<code>=(C4-(B4*C2+C3))/B4</code>								
5	4Li/	4	-41177.38125	-1.681467499998								
6	8Li/	8	-41966.63027	-1.823151249998								
7	16Li/	16	-43539.61856	-1.549633749998								
8	32Li/	32	-46688.68162	-1.509327499998								
9												
10												

[Lab.3] Adsorption energies of Li atoms for different coverage

11. Calculate adsorption energies

$$E_{adsorption} = -[E_{adsorption\ structure} - (n_{Li-atoms} \times E_{Li-atom} + E_{MoS_2})] / n_{Li-atoms}$$



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Thank you

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